# Calculations for the preferred design (Design A): beam and leg seat set

## 1. Assumptions

- Sizes and masses are taken from the CAD model (models/esp2026.py): every one of the 39 parts is modelled with its own material density. One set has 128 parts.
- Life vests are supplied and replaced by the airline (they are cabin safety equipment), so they are left out of the 8 kg seat limit. They are listed with them included as well.
- Seat pitch 31 in (787 mm), typical for economy. A standard passenger is 77 kg (CS 25.562). Forward emergency landing load 9 g (CS 25.561), checked as a static load on the legs.
- Labour £15 an hour (brief). Material prices, labour times and bought in prices are estimates from supplier catalogues for a batch of 100 sets; they are listed in section 8.
- Tooling (moulds, jigs) £60,000 and certification testing (16 g sled tests, flammability) £150,000 are shared over the 100 sets.
- Cabin 18 to 27 °C: aluminium, carbon fibre and the polymers chosen are all stable well beyond this range, so no thermal allowance is needed.
- “Sustainable” means recycled content, bio based, or recycled at end of life through an established route (aluminium, steel, thermoplastics, wool and PET).

## 2. Weight (mass)

| Quantity | Working | Result |
|---|---|---|
| Mass of one three-seat set (CAD model, no life vests) | sum of volume × density for all 128 parts | **23.50 kg** |
| Mass per seat | 23.50 / 3 | **7.83 kg** |
| Limit (brief): frame, cushions and belts | 8 kg a seat, so margin = 8 − 7.83 | **0.17 kg under** |
| With the three life vests (for information) | 25.04 / 3 | 8.35 kg |

| Group (one set) | Mass |
|---|---|
| Structure | 7.67 kg |
| Seat | 8.44 kg |
| Recline | 0.94 kg |
| Armrest | 2.00 kg |
| Belt | 1.05 kg |
| Amenity | 3.41 kg |

The structure (beam tubes, legs, spreaders, track fittings) is shared by the three seats. The heaviest single items per seat are the carbon back shell (0.82 kg) and the cushion foams. The back shell is only 1.2 mm thick: carbon fibre is used because it is about 40 % lighter than an aluminium back of the same stiffness.

## 3. Size

| Quantity | Working | Result |
|---|---|---|
| Overall width of the set | CAD bounding box: beam tubes and end caps | **1554 mm** |
| Seat width between armrests | seat pitch across 500 − armrest 50 (brief 432 to 457) | **450 mm** |
| Overall depth, backs upright | front of cushion to back of the shell | **709 mm** |
| Overall height | floor to the top of the head rest (the track studs go 20 mm into the floor) | **1181 mm** |
| Back moves rearwards when reclined 5° | CAD, reclined minus upright | 63 mm |
| Overall depth, backs reclined | less than the 787 mm pitch, so the set stays in its own row | **772 mm** |
| Track fitting centres | 20 track holes at 25.4 mm | 508 mm |
| Tray table (brief up to 267 × 419) | CAD | **265 × 415 mm** |
| Life vest pouch inside (vest 305 × 305 × 50) | CAD | **316 × 336 × 56 mm** |

## 4. Seat height and ergonomics

| Quantity | Working | Result |
|---|---|---|
| Seat height (front of the cushion) | beam axis 330 + tube radius 25 + diaphragm and foam | **440 mm** |
| Target from body sizes (popliteal height + shoe) | 5th % woman 355 + 25 shoe = 380; 95th % man 490 + 25 = 515 | 380 to 515 |
| Cushion compresses under a passenger | about 30 mm, so the sitting height is | 410 mm |
| Armrest top above the floor | 640 + pad 8 (elbow rest about 200 above the seat) | 648 mm |
| Cushion depth | 370 − (-80) (5th % woman buttock to knee 520) | 450 mm |
| Back angle upright / reclined | 12° + 5° recline (brief 3 to 6°) | **12° / 17°** |

Most passengers can sit with their feet flat; the 3.8° cushion slope stops sliding forward; lifting armrests allow a sideways transfer from an aisle wheelchair.

## 5. Cost to manufacture

| Cost element (one three-seat set) | Working | Result |
|---|---|---|
| Raw materials | finished mass × buy to fly ratio × £/kg, + 5 % scrap (table section 8) | £855 |
| Bought in parts | track studs, recline locks, belt buckles, screens, power modules | £1,257 |
| Labour | 51.0 h making parts + 8 h assembly = 59.0 h × £15 | £885 |
| Factory overhead | 30 % of labour (power, machines, floor space) | £266 |
| Tooling share | £60,000 / 100 sets | £600 |
| Certification testing share | £150,000 / 100 sets | £1,500 |
| **Cost to manufacture one set** | | **£5,362** |
| Budget (brief) | £1.5 million / 100 sets | £15,000 |
| **Margin for profit, delivery and fitting** | **£15,000 − £5,362** | **£9,638** |
| Whole order | £5,362 × 100 | £536,207 |
| Time: workers needed for 100 sets in 26 weeks | 59.0 h × 100 / (26 × 37.5 h) at 80 % efficiency | 7.6 people |

The cost is well inside the budget, so the airline can be offered a competitive price. The biggest single cost is the screens (£660 a set); a cheaper screen or a phone holder would save most. Six to eight weeks are allowed for testing and certification at the start of the 26 weeks, then about 10 people make the 100 sets in the remaining 19 weeks (the legs and spreaders are machined in batches).

## 6. Strength: the legs in a 9 g emergency landing

Forward load on the set: 9 × 9.81 × (3 × 77 + 23.5) = 22,470 N, shared by two legs: **11,235 N a leg**, half at each beam bore. The leg is treated as a pin jointed truss (A, B feet; C, D beam bores). Joint equilibrium gives the member forces; stress = force / (strut width × 10 mm plate). Foot reactions: front 6856 N back, 6303 N up; rear 4379 N back, 6303 N down (the rear foot is pulled up).

| Member | Force | Section (mm) | Hand stress | FEA (mid strut) |
|---|---|---|---|---|
| front strut | 0 N tension | 40 × 10 | 0.0 MPa | 2.5 MPa |
| rear strut | 7675 N tension | 40 × 10 | 19.2 MPa | 18.7 MPa |
| top chord | 5618 N tension | 30 × 10 | 18.7 MPa | 18.0 MPa |
| brace | 9313 N compression | 30 × 10 | 31.0 MPa | 28.8 MPa |

The hand calculation and the FEA (gmsh + scikit-fem, 10 node tetrahedra) agree within 7 % in the main members; the front strut carries no load in a pure truss and only a little bending in the FEA. The FEA peak is **227 MPa** in the corner where the brace meets the rear boss: safety factor **2.2** on 7075-T6 (yield 503 MPa). Largest movement 0.51 mm. **OK.** Rear foot uplift 6303 N is taken by the shear plunger track stud (rated about 13 kN).

## 7. Recline mechanism and seat belt

| Quantity | Working | Result |
|---|---|---|
| Lock pin centres, upright / reclined | from the pivot geometry (lever -100 mm below the pivot) | 353.9 / 345.1 mm |
| Lock stroke used | 353.9 − 345.1 | **8.8 mm** |
| Lever arm of the lock about the pivot | perpendicular distance, pivot to lock line | 100 mm |
| Lock force, passenger leaning back | 300 N at 450 mm: 300 × 450 / 100 | 1350 N (rated 1500 N) |
| Belt length | buckle side 380 + adjuster side 790 to 1040 | **1170 to 1420 mm** |

## 8. Materials, prices and sustainability

| Material | £/kg | Buy to fly | Sustainable | Why |
|---|---|---|---|---|
| 7075-T6 aluminium plate, machined | £12 | 5.0 | 100 % | Recycled at end of life (aluminium keeps its value) |
| 7075/6061 aluminium tube | £9 | 1.2 | 100 % | 70 % recycled content, recyclable |
| 17-4PH stainless steel bar | £15 | 3.0 | 100 % | Recyclable steel |
| FR polycarbonate, moulded | £6 | 1.1 | 100 % | Thermoplastic, recyclable |
| Recycled PA6 GF30, moulded | £5 | 1.1 | 100 % | Recycled content |
| TPE over foam | £6 | 1.2 | 0 % | Not counted |
| Carbon fibre prepreg | £60 | 1.5 | 0 % | Thermoset: not counted |
| FR polyurethane foam | £8 | 1.1 | 30 % | 30 % bio based polyol |
| Wool and recycled polyester cover | £30 | 1.3 | 100 % | Natural and recycled fibre |
| Aramid and polyester suspension | £40 | 1.2 | 50 % | Polyester share recycled |
| Recycled PET felt, moulded | £6 | 1.2 | 100 % | Recycled content |
| Al honeycomb sandwich, PC faces | £50 | 1.3 | 100 % | Aluminium and PC, recyclable |
| Polyester webbing (safety) | £15 | 1.1 | 0 % | Virgin fibre for strength: not counted |
| Foam on PA6 frame, recycled leather | £20 | 1.2 | 60 % | PA6 frame and recycled leather fibre |

Sustainable share by mass (no life vests): 16.77 kg of 23.50 kg = **71 %** (brief: at least 70 %). **OK.** The share only just passes. The back shells are the largest item not counted (2.46 kg a set): made from recycled carbon fibre they would raise it to about 82 %.

## 9. Cost, mass and sustainability of every part (one set)

| Item | Part | Qty | Mass kg | Material | Material | Hours | Bought in | Sust. |
|---|---|---|---|---|---|---|---|---|
| 1 | beam tube | 2 | 1.978 | 7075/6061 aluminium tube | £22 | 1.00 | – | 100 % |
| 2 | beam end cap | 4 | 0.156 | FR polycarbonate, moulded | £1 | 0.20 | – | 100 % |
| 3 | spreader | 4 | 1.934 | 7075-T6 aluminium plate, machined | £122 | 6.00 | – | 100 % |
| 4 | leg | 2 | 2.646 | 7075-T6 aluminium plate, machined | £167 | 6.00 | – | 100 % |
| 5 | beam clamp collar | 4 | 0.122 | 7075-T6 aluminium plate, machined | £8 | 0.80 | – | 100 % |
| 6 | track fitting | 4 | 0.343 | 7075-T6 aluminium plate, machined | £22 | 2.00 | – | 100 % |
| 7 | track stud | 4 | 0.210 | Bought in | – | 0.20 | £72 | 100 % |
| 8 | baggage bar | 1 | 0.279 | 7075/6061 aluminium tube | £3 | 0.30 | – | 100 % |
| 9 | seat diaphragm | 3 | 0.572 | Aramid and polyester suspension | £29 | 1.50 | – | 50 % |
| 10 | seat cushion foam | 3 | 1.786 | FR polyurethane foam | £17 | 0.90 | – | 30 % |
| 11 | seat cushion cover | 3 | 0.579 | Wool and recycled polyester cover | £24 | 3.00 | – | 100 % |
| 12 | back shell | 3 | 2.462 | Carbon fibre prepreg | £233 | 7.50 | – | 0 % |
| 13 | back cushion foam | 3 | 1.163 | FR polyurethane foam | £11 | 0.90 | – | 30 % |
| 14 | back cushion cover | 3 | 0.641 | Wool and recycled polyester cover | £26 | 3.00 | – | 100 % |
| 15 | head rest | 3 | 1.009 | Foam on PA6 frame, recycled leather | £25 | 2.40 | – | 60 % |
| 16 | recline pivot pin | 2 | 0.140 | 17-4PH stainless steel bar | £7 | 0.20 | – | 100 % |
| 17 | end recline pivot pin | 2 | 0.087 | 17-4PH stainless steel bar | £4 | 0.20 | – | 100 % |
| 18 | recline lock bracket | 3 | 0.136 | 7075-T6 aluminium plate, machined | £9 | 0.90 | – | 100 % |
| 19 | recline lock body | 3 | 0.484 | Bought in | – | 0.30 | £255 | 100 % |
| 20 | recline lock rod | 3 | 0.256 | Bought in | – | 0.00 | – | 100 % |
| 21 | recline lock pin | 6 | 0.060 | 17-4PH stainless steel bar | £3 | 0.60 | – | 100 % |
| 22 | recline button | 3 | 0.008 | FR polycarbonate, moulded | £0 | 0.15 | – | 100 % |
| 23 | armrest | 4 | 1.483 | Recycled PA6 GF30, moulded | £9 | 0.80 | – | 100 % |
| 24 | armrest pad | 4 | 0.311 | TPE over foam | £2 | 0.80 | – | 0 % |
| 25 | armrest pivot pin | 4 | 0.202 | 17-4PH stainless steel bar | £10 | 0.40 | – | 100 % |
| 26 | belt anchor | 6 | 0.359 | 7075-T6 aluminium plate, machined | £23 | 1.80 | – | 100 % |
| 27 | belt webbing, buckle side | 3 | 0.069 | Polyester webbing (safety) | £1 | 0.60 | – | 0 % |
| 28 | belt webbing, adjuster side | 3 | 0.069 | Polyester webbing (safety) | £1 | 0.60 | – | 0 % |
| 29 | buckle body | 3 | 0.268 | Bought in | – | 0.15 | £75 | 100 % |
| 30 | buckle lift lever | 3 | 0.092 | Bought in | – | 0.00 | £15 | 100 % |
| 31 | belt tongue | 3 | 0.190 | Bought in | – | 0.15 | £45 | 100 % |
| 32 | tray table | 3 | 0.982 | Al honeycomb sandwich, PC faces | £67 | 3.00 | – | 100 % |
| 33 | tray hinge arm | 6 | 0.080 | 7075-T6 aluminium plate, machined | £5 | 1.20 | – | 100 % |
| 34 | tray latch | 3 | 0.012 | FR polycarbonate, moulded | £0 | 0.15 | – | 100 % |
| 35 | entertainment screen | 3 | 1.413 | Bought in | – | 0.90 | £660 | 30 % |
| 36 | power and audio module | 3 | 0.106 | Bought in | – | 0.60 | £135 | 30 % |
| 37 | literature pocket | 3 | 0.284 | Recycled PET felt, moulded | £2 | 0.90 | – | 100 % |
| 38 | life vest pouch | 3 | 0.531 | Recycled PET felt, moulded | £4 | 0.90 | – | 100 % |
| 39 | life vest | 3 | 1.535 | Bought in | – | 0.00 | – | excl. |